

ASSIGNMENT 2

1. Explain the responsibilities of the data-link layer and distinguish logical link control from media access control.

ANS:

Responsibilities of the Data-Link Layer

1. **Framing**

The data-link layer organizes the network-layer packet into a **frame** by adding a header and, depending on the protocol, a trailer.

2. **Physical/MAC addressing**

It uses MAC addresses to identify devices on a local network.

3. **Error detection**

It can detect transmission errors using mechanisms such as the Frame Check Sequence (FCS), normally based on CRC.

4. **Media access control**

When multiple devices share a communication medium, the data-link layer determines how they access that medium.

5. **Flow control**

Some data-link protocols provide mechanisms to prevent a fast sender from overwhelming a receiver.

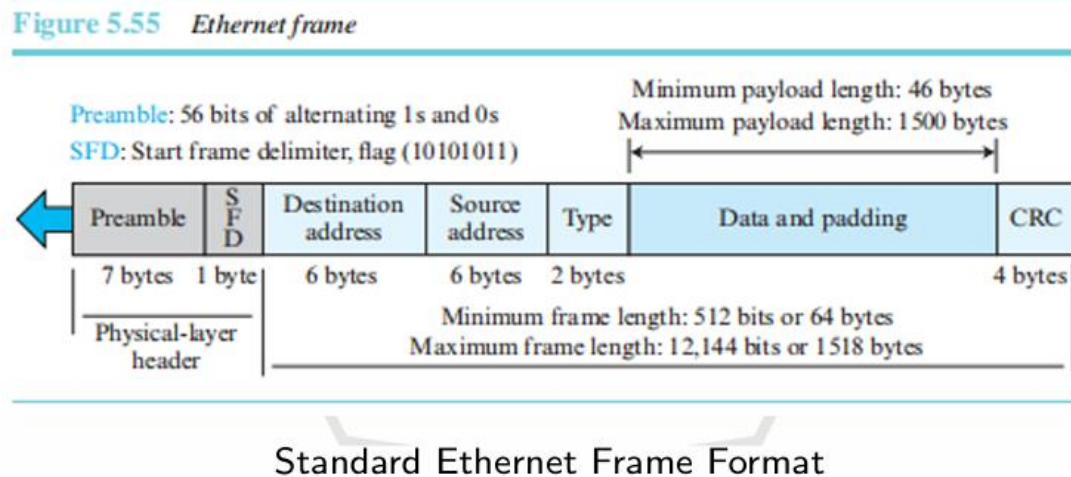
6. **Link management**

Data-link protocols can provide functions for establishing, maintaining, and managing communication over a link.

LOGICAL LINK CONTROL	MEDIA ACCESS CONTROL
Logical Link Control	Media Access Control
Upper sublayer	Lower sublayer
Provides an interface between the network layer and MAC sublayer	Controls access to the transmission medium
Defined by IEEE 802.2	Defined separately for technologies such as Ethernet and IEEE 802.11
Can identify/multiplex network-layer protocols	Provides MAC addressing and frame-related functions

2. Explain Ethernet frame format and identify the purpose of destination MAC, source MAC, type/length and FCS fields.

ANS:



1. Preamble – 7 bytes

The preamble consists of a pattern used by the receiver for synchronization before the frame begins.

2. Start Frame Delimiter (SFD) – 1 byte

The SFD indicates that the actual Ethernet frame is about to begin.

3. Destination MAC Address – 6 bytes

Identifies the intended destination at the data-link layer.

Example:

00:1A:2B:3C:4D:5E

A destination can be:

- Unicast – one device
- Multicast – a group of devices
- Broadcast – all devices on the LAN

Broadcast MAC:

FF:FF:FF:FF:FF:FF

4. Source MAC Address – 6 bytes

Identifies the device that transmitted the frame.

5. EtherType/Length – 2 bytes

In an Ethernet II frame, this field is called EtherType and identifies the protocol carried in the payload.

Examples:

EtherType Protocol

0x0800 IPv4

0x0806 ARP

0x86DD IPv6

In an IEEE 802.3 frame, the corresponding 2-byte field can indicate the length of the MAC client data rather than an EtherType.

6. Data/Payload

Contains the packet received from the network layer. For standard Ethernet, the MAC data field is normally 46 to 1500 bytes. Padding may be added when necessary to satisfy the minimum Ethernet frame size.

7. FCS – 4 bytes

FCS = Frame Check Sequence

It contains a CRC-based value used by the receiver to detect errors in the received frame. If an error is detected, the corrupted frame is normally discarded.

3. Use Wireshark to capture Ethernet traffic and identify source and destination MAC addresses from at least five frames

ANS:

4. Explain the operation of a switch using MAC address learning. Demonstrate how a switch builds its forwarding table

ANS:

The Two Core Mechanisms of a Switch

1. MAC Address Learning (Source Inspection):

When a frame enters any port, the switch examines the frame's Source MAC Address and the port it arrived on. If that MAC address is not already in its table, the switch records the MAC address along with the corresponding port number, binding them together with an aging timer.

2. Frame Forwarding & Filtering (Destination Lookup):

The switch then looks at the frame's Destination MAC Address and checks its MAC address table:

- Known Unicast: If the destination MAC address is found in the table, the switch forwards the frame only out of the specific port listed. It filters (drops) the frame from going out of other ports, reducing network congestion.
- Unknown Unicast / Broadcast: If the destination MAC address is not in the table (or if it is a broadcast address like FF:FF:FF:FF:FF:FF), the switch performs flooding, sending the frame out of all other ports except the one it arrived on.

Step-by-Step Demonstration: Building the Forwarding Table

Imagine a small switch with three hosts connected to its ports:

- Host A (MAC: AA:AA:AA:AA:AA:AA) connected to Port 1
- Host B (MAC: BB:BB:BB:BB:BB:BB) connected to Port 2
- Host C (MAC: CC:CC:CC:CC:CC:CC) connected to Port 3

Initially, the switch's MAC address table is completely empty.

Step 1: Host A sends data to Host B

- Action: Host A (AA:AA:AA...) transmits a frame destined for Host B (BB:BB:BB...) into Port 1.
- Learning: The switch checks the source MAC address (AA:AA:AA...) and records: AA:AA:AA... is on Port 1.
- Forwarding: The switch looks up the destination MAC (BB:BB:BB...). Because the table doesn't have BB:BB:BB... yet, the switch floods the frame out of all other ports (Port 2 and Port 3).
- Result: Host B receives the frame and processes it. Host C receives it, sees the destination MAC doesn't match its own, and drops it.

Step 2: Host B replies to Host A

- Action: Host B (BB:BB:BB...) replies, sending a frame destined for Host A (AA:AA:AA...) into Port 2.
- Learning: The switch checks the source MAC (BB:BB:BB...) and records: BB:BB:BB... is on Port 2.
- Forwarding: The switch looks up the destination MAC (AA:AA:AA...). It checks its table and finds AA:AA:AA... is associated with Port 1.

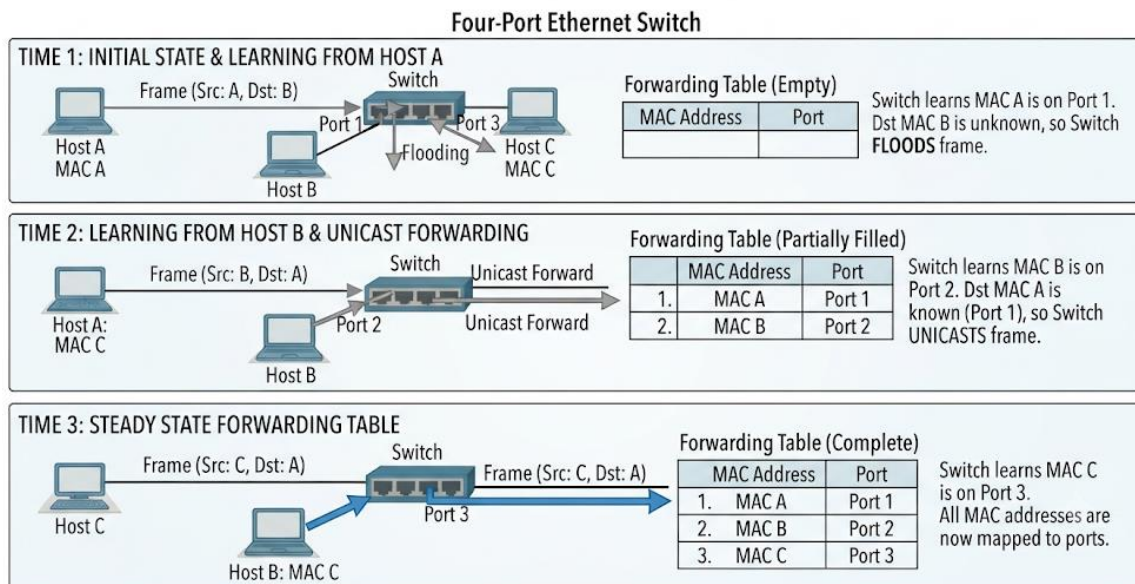
- Result: The switch unicasts the frame directly and exclusively out of Port 1. No other ports receive unnecessary traffic.

Step 3: Host C communicates

If Host C (CC:CC:CC...) eventually sends a frame out of Port 3, the switch adds a third entry (CC:CC:CC... on Port 3).

Over time, all active devices are learned, enabling the switch to handle multiple simultaneous point-to-point conversations without collisions or wasted bandwidth. Entries that remain inactive for a specified period (aging timer, usually 300 seconds) are automatically flushed from the table to accommodate network changes and mobile devices.

DYNAMIC MAC ADDRESS LEARNING AND FORWARDING TABLE CONSTRUCTION



5. Compare hub, switch and bridge in terms of forwarding, collision domains and practical use

ANS:

Feature	Hub	Bridge	Switch
OSI layer	Layer 1	Layer 2	Layer 2
Uses MAC addresses	No	Yes	Yes
Forwarding decision	None; repeats signal	Based on MAC address	Based on MAC address

Feature	Hub	Bridge	Switch
Ports	Usually multiple	Usually few	Usually many
Collision domains	One shared collision domain	Separates collision domains	Each switch port is normally a separate collision domain
Broadcast domain	Does not separate it	Does not normally separate it	Does not normally separate it
Duplex	Typically half-duplex	Can support full-duplex	Commonly full-duplex
Current practical use	Mostly obsolete	Largely replaced by switches	Widely used

6. Explain CSMA/CD and discuss why its importance is reduced in modern switched full-duplex Ethernet

ANS:

CSMA/CD stands for: Carrier Sense Multiple Access with Collision Detection

It was used in traditional shared, half-duplex Ethernet networks.

Operation

1. Carrier Sense
A device listens to determine whether the medium is busy.
2. Multiple Access
Multiple devices share the same transmission medium.
3. Transmit
If the medium is idle, a device begins transmission.
4. Collision Detection
If two devices transmit simultaneously, a collision can occur. Devices detect the collision.
5. Backoff
The devices stop transmitting and wait for a calculated random backoff period.
6. Retransmission
After the backoff period, the devices attempt transmission again.

Why is CSMA/CD less important now?

Traditional shared Ethernet used hubs or shared coaxial media, where multiple devices competed for the same medium.

Modern Ethernet networks normally use:

Device $\longleftrightarrow$ Switch

with full-duplex links.

In full-duplex Ethernet:

- A device can transmit and receive simultaneously.
- The link is point-to-point.
- There is no shared collision domain on that link.
- Therefore, collisions do not occur during normal operation.

Consequently, CSMA/CD is not used for normal full-duplex switched Ethernet operation.

7. Explain CSMA/CA in IEEE 802.11 and compare its objective with CSMA/CD

ANS:

CSMA/CA stands for: Carrier Sense Multiple Access with Collision Avoidance

It is used by IEEE 802.11 wireless LANs.

Wireless stations share a radio channel, so a mechanism is needed to coordinate access.

Basic CSMA/CA operation

1. A station senses the wireless channel.
2. If the channel is busy, it waits.
3. If the channel remains idle for the required interframe space, the station uses a random backoff counter.
4. The counter decreases while the channel remains idle.
5. When the counter reaches zero, the station transmits.
6. The receiver can respond with an ACK.
7. If the expected ACK is not received, the sender assumes the transmission was unsuccessful and may retransmit using contention/backoff procedures.

IEEE 802.11 can also use RTS/CTS in appropriate situations, such as helping address the hidden-node problem.

Why avoidance rather than detection?

A wireless station generally cannot reliably listen for a collision while it is transmitting because its own transmitted signal is much stronger than a received signal. Therefore, Wi-Fi focuses on collision avoidance, rather than Ethernet's traditional collision-detection method.

Comparison

CSMA/CD	CSMA/CA
Traditional shared Ethernet	IEEE 802.11 WLAN
Collision Detection	Collision Avoidance
Detects collision during transmission	Attempts to reduce the probability of collision
Used with shared half-duplex Ethernet	Used for wireless medium access
No ACK mechanism equivalent to 802.11 MAC ACK for every ordinary Ethernet frame	MAC ACK commonly used for unicast data frames
Modern switched full-duplex Ethernet does not normally use it	Still fundamental to 802.11 channel access

8. Design a small wireless LAN for a college laboratory. Specify access point placement, SSID, channel considerations, addressing and basic security requirements

ANS:

A small wireless LAN can be designed to provide reliable and secure Wi-Fi access to students and faculty in a college laboratory.

1. Access Point Placement

- Place one wireless access point (AP) near the centre of the laboratory.
- Keep it at a suitable height and away from walls, metal objects and electrical interference.
- If the laboratory is large, use two APs to provide better coverage.
- Avoid placing the AP inside a cupboard or near the floor.

2. SSID

- Create a unique SSID such as College_Lab_WiFi.

- The SSID should be easily identifiable but should not contain sensitive information.
- A separate SSID can be created for students and faculty if required.

3. Channel Considerations

- For the 2.4 GHz band, use non-overlapping channels such as 1, 6 or 11.
- Avoid using the same channel for nearby APs to reduce interference.
- If supported, use the 5 GHz band because it generally provides more channels and less interference.
- Select the channel based on a Wi-Fi scan of nearby networks.

4. IP Addressing

Example private network:

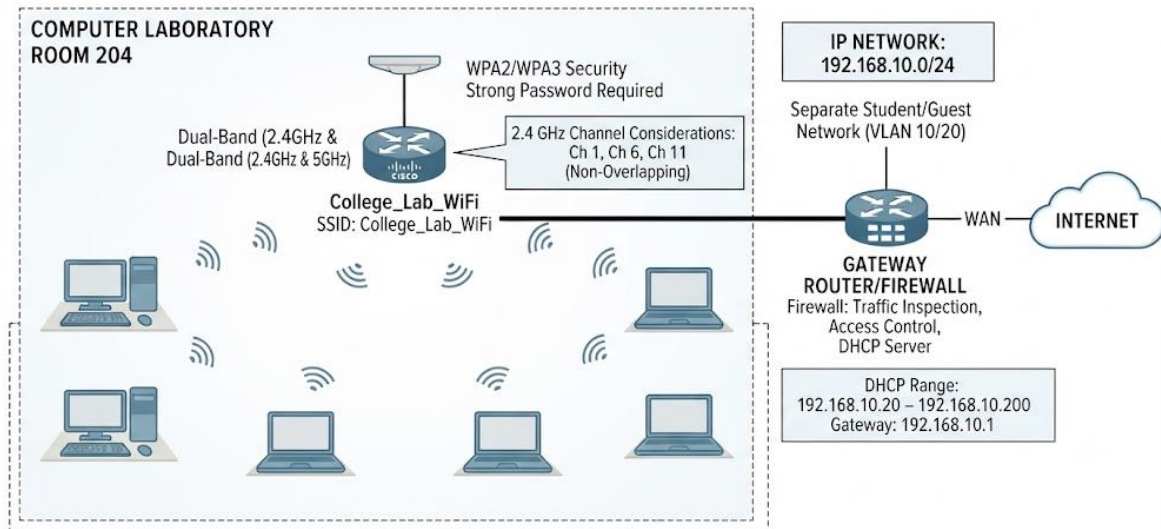
- Network: 192.168.10.0/24
- AP/Router: 192.168.10.1
- DHCP range: 192.168.10.20 – 192.168.10.200
- Subnet mask: 255.255.255.0
- Gateway: 192.168.10.1
- DNS: Provided by the router or institution.

The DHCP server automatically assigns IP addresses to student laptops and other devices.

5. Basic Security Requirements

- Use WPA2-AES or preferably WPA3 security.
- Set a strong Wi-Fi password.
- Change the default administrator username/password of the AP.
- Disable WPS if it is not required.
- Keep the AP firmware updated.
- Use a firewall on the router.
- Create a separate guest/student network if possible.
- Monitor connected devices and remove unknown devices.

WIRELESS LAN NETWORK TOPOLOGY: COLLEGE COMPUTER LABORATORY



9. Explain Mobile IP and the need for mobility support when a device moves between networks

ANS:

Mobile IP is a network-layer mobility mechanism designed to allow a mobile device to remain reachable while changing its point of attachment to the network.

Why mobility support is required

An IP address normally has a relationship with the network where it is assigned.

Suppose a device moves from:

Home Network



Visited Network

If the device simply changes to a new IP address, existing communication sessions may be disrupted because packets addressed to its original address are still associated with the home network.

Mobile IP provides a mechanism for maintaining reachability while the device is away from its home network.

Mobile IP components

1. Mobile Node (MN)

The device that moves between networks.

2. Home Agent (HA)

A router on the mobile node's home network that maintains information about the mobile node's current location.

3. Foreign Agent (FA)

A Mobile IPv4 entity on a visited network that can provide a care-of address and mobility-related services to the mobile node.

Care-of Address

When the mobile node moves to another network, it obtains a care-of address (CoA) representing its current location.

The Home Agent can tunnel packets destined for the mobile node's home address toward the CoA.

Purpose

Mobile IP allows communication to continue while the device changes its network attachment point, without requiring the correspondent node to know the mobile node's temporary location.

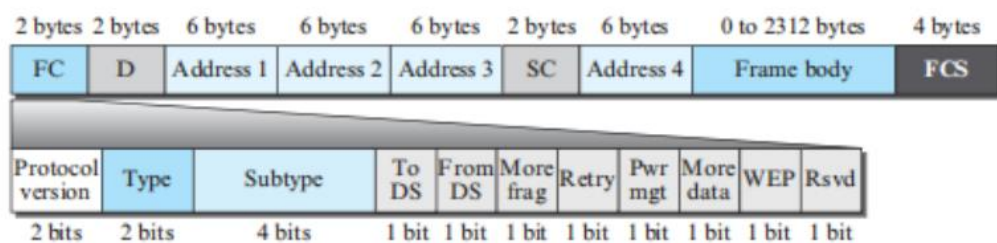
10. Perform a packet-level comparison of a wired Ethernet frame and a wireless LAN frame using suitable captures or documented examples

ANS:

Feature	Wired Ethernet Frame	Wireless LAN Frame
Standard	IEEE 802.3	IEEE 802.11
Transmission medium	Ethernet cable	Radio waves
MAC addresses	Usually 2 MAC addresses	Can contain up to 4 MAC addresses
Main address fields	Destination, Source	Receiver, Transmitter, Destination, sometimes Source
Type/Length	Present	Not directly used like Ethernet; frame control and other fields are used
Frame Control	Not present	Present
Sequence Control	Not present	Present

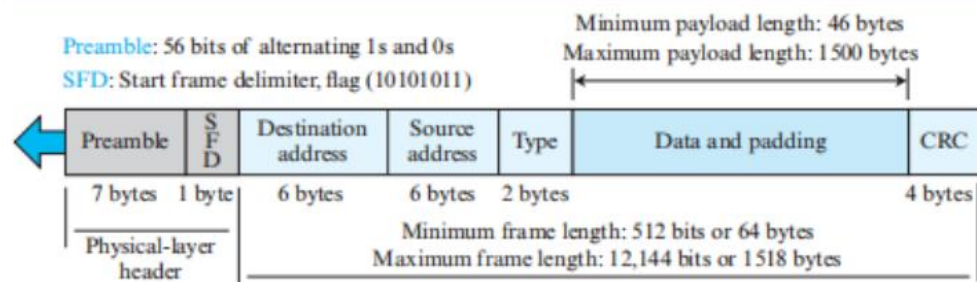
Feature	Wired Ethernet Frame	Wireless LAN Frame
FCS	Present	Present
Error/medium handling	Mainly Ethernet mechanisms	Designed for wireless interference and mobility
Typical overhead	Lower	Higher

Figure 6.11 Frame format



Wireless Lan Frame Format

Figure 5.55 Ethernet frame



Standard Ethernet Frame Format